What a Neural Jastrow Factor Can and Cannot Learn: A Controlled Variational Monte Carlo Study of the Beryllium Atom

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We study the ground state of the beryllium atom with variational Monte Carlo (VMC) and fixed-node diffusion Monte Carlo (FN-DMC). The trial wave functions are products of a determinant part and a Jastrow factor that contains a neural network. The Jastrow factor is positive, so FN-DMC measures the quality of the determinant part alone. In a series of four single-determinant trial functions with the same node, a neural network must learn the electron-electron cusps. It reproduces the antiparallel-spin cusp to 0.7% and misses the parallel-spin cusp by 81%, with no measurable effect on the energy. For a complete-active-space expansion over the $2s^2$ and $2p^2$ configurations we show that the determinant part factorises into a positive function and a nodal function of two radial functions and one number, and we give the transformations of the orbitals that leave the node unchanged. Across nine trial functions, those with a radial modulation of the $1s$ orbital have a fixed-node energy 0.20(3) mHa lower than those without it. Neither the VMC energy nor the configuration-mixing coefficient detects this difference, and a single gauge-invariant number does not order nodes whose radial functions differ. A $3d^2$ configuration and a one-body Jastrow term do not change the node measurably. The difference $E_{\mathrm{VMC}} - E_{\mathrm{FN}}$ is 0.2–0.8 mHa, larger than the fixed-node error, which is 0.14(2) mHa for the trial functions with a modulated $1s$ orbital.

I. INTRODUCTION

A common trial wave function for electrons has the Slater-Jastrow form,

$$\Psi(\mathbf{R}) = D(\mathbf{R})\, e^{J(\mathbf{R})}. \tag{1}$$

The determinant part D is antisymmetric. The Jastrow factor e^J is symmetric and positive, and describes the correlation of electron pairs at short distance. Since $e^J > 0$, the nodal surface of Ψ, the set where $\Psi = 0$, is the nodal surface of D. Neural networks are now used to represent J and, in some approaches, the whole wave function [1–3]. A network inside J increases the flexibility of the positive factor. It does not change the node.

The variational Monte Carlo (VMC) energy contains two errors: the error of the node and the error of the positive factor. Fixed-node diffusion Monte Carlo (FN-DMC) separates them. In the limit of zero time step it gives the lowest energy compatible with the node of the trial function, whatever the positive factor [4, 5]. The difference $E_{\mathrm{VMC}} - E_{\mathrm{FN}}$ is then the error of the positive factor, and $E_{\mathrm{FN}} - E_{\mathrm{exact}}$ is the error of the node. When two trial functions differ in both factors, their VMC energies do not tell which factor is responsible for the difference.

The beryllium atom is a test case for this question. Its $2s^2$ configuration is nearly degenerate with $2p^2$, and a single determinant gives a fixed-node error of order 10 mHa [6]. The exact non-relativistic energy is known to better than $1\,\mu E_h$ [7]. With four electrons the node of a few-configuration wave function can be written in closed form [8].

We study two sets of trial functions. The first is a series of four single-determinant trial functions with the same node and different Jastrow factors, from none to a neural network that must learn the electron-electron cusps. The second is a complete-active-space expansion over $2s^2$ and $2p^2$, with optional $3d^2$ configurations, multiplied by a Jastrow factor with a three-body neural network. For the second set we show that the determinant part factorises into a positive function and a nodal function, and we list the transformations of the orbitals that leave the node unchanged. We then compute VMC and FN-DMC energies for nine trial functions that differ in one element at a time: the radial form of the orbitals, a one-body Jastrow term, the $3d^2$ configurations, and the random seed of the optimisation.

The results are the following. In the single-determinant series the network learns the antiparallel-spin cusp and not the parallel-spin cusp, and the energy does not depend on the latter. The VMC error of the multi-configuration trial functions comes mostly from the Jastrow factor. The node depends on the shape of the $1s$ orbital: the trial functions in which the $1s$ orbital is modulated have a fixed-node energy 0.20(3) mHa lower than the others. Neither the VMC energy nor the mixing coefficient of the $2p^2$ configurations shows this. The mixing coefficient is not identifiable from the optimisation, and its gauge-invariant version does not order nodes whose radial functions differ. The $3d^2$ configurations and a one-body Jastrow term do not change the node measurably.

Section II defines the trial functions. Section III derives the structure of the nodal surface. Section IV describes the VMC and FN-DMC calculations. Section V presents the results, Sec. VI discusses them, and Sec. VII states the limitations of the study.

II. TRIAL WAVE FUNCTIONS

A. Hamiltonian and antisymmetry

In Hartree atomic units the Hamiltonian of the beryllium atom with a fixed nucleus of charge $Z = 4$ is

$$\hat{H} = -\frac{1}{2}\sum_{i=0}^{3}\nabla_i^2 - \sum_{i=0}^{3}\frac{Z}{r_i} + \sum_{0\le i<j\le 3}\frac{1}{r_{ij}}, \tag{2}$$

with $r_i = |\mathbf{r}_i|$ and $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. $\hat{H}$ does not act on spin. We fix $S_z = 0$, with electrons $0, 1$ of spin up and $2, 3$ of spin down. For two spatial orbitals ϕ_a, ϕ_b the determinant part is the product of two 2×2 determinants,

$$D[\phi_a, \phi_b] = \det M_\uparrow \, \det M_\downarrow, \tag{3}$$

with $M_\uparrow = \big(\phi_k(\mathbf{r}_i)\big)$ for $i \in \{0, 1\}$ and $M_\downarrow = \big(\phi_k(\mathbf{r}_i)\big)$ for $i \in \{2, 3\}$, $k \in \{a, b\}$. It changes sign under the exchange of two electrons of the same spin. Of the six electron pairs, two have parallel spins, $(0, 1)$ and $(2, 3)$, and four antiparallel spins.

B. Orbitals

When an electron approaches the nucleus, $-Z/r_i$ diverges. A finite local energy requires, for an s orbital, the electron-nucleus cusp condition [9]

$$\left.\frac{\phi'(r)}{\phi(r)}\right|_{r\to 0} = -Z. \tag{4}$$

We impose it exactly. For $\phi_{1s} \propto e^{-\alpha_{1s} r}$ it gives $\alpha_{1s} = Z$, which we keep fixed. For $\phi_{2s} \propto (r - c)\, e^{-\alpha_{2s} r}$ one has $\phi(0) = -c$ and $\phi'(0) = 1 + \alpha_{2s} c$, so Eq. (4) fixes the position of the radial node,

$$c = \frac{1}{Z - \alpha_{2s}}. \tag{5}$$

c is recomputed from α_{2s} at every evaluation. The $2p$ and $3d$ orbitals vanish at the nucleus and carry no cusp condition.

Each orbital is multiplied by a radial modulation built on the coordinate

$$t(r) = \frac{r^2}{1 + r}, \qquad t(0) = 0, \quad t'(0) = 0. \tag{6}$$

The orbitals are

$$\phi_{1s}(r) = e^{-Zr}\, m_{1s}(r), \tag{7}$$

$$\phi_{2s}(r) = (r - c)\, e^{-\alpha_{2s} r}\, m_{2s}(r), \tag{8}$$

$$\phi_{2p_k}(\mathbf{r}) = x_k\, e^{-\alpha_{2p} r}\, m_{2p}(r), \qquad k \in \{x, y, z\}, \tag{9}$$

and, for the $3d$ shell, the five components of $T(\mathbf{r})\, e^{-\alpha_{3d} r}\, m_{3d}(r)$, with $T_{ab}(\mathbf{r}) = x_a x_b - \delta_{ab} r^2/3$. The modulation is

$$m_{nl}(r) = 1 + c_{nl}^{(1)}\, t(r) + c_{nl}^{(2)}\, t(r)^2. \tag{10}$$

Since $t(0) = t'(0) = 0$, m_{nl} has unit value and zero slope at the nucleus, and Eq. (4) holds for any value of the coefficients. The coefficients are bounded, $c^{(1)} \in [-0.15, 2.5]$ and $c^{(2)} \in [-0.15, 0.8]$, and the exponents are bounded, $\alpha_{2s}, \alpha_{2p} \in [0.3, 3]$ and $\alpha_{3d} \in [0.3, 5]$. A parameter that reaches a bound has zero gradient and stays there; we report which parameters do.

C. Determinant part

The single-determinant trial functions use $D[\phi_{1s}, \phi_{2s}]$, with linear modulations, $c_{1s}^{(2)} = c_{2s}^{(2)} = 0$, and three variational parameters, α_{2s}, $c_{1s}^{(1)}$ and $c_{2s}^{(1)}$.

The multi-configuration trial functions use

$$\begin{aligned} D_{\mathrm{CAS}} = D[\phi_{1s}, \phi_{2s}] + c_p \sum_k D[\phi_{1s}, \phi_{2p_k}] \\ + c_d \sum_m D[\phi_{1s}, \phi_{3d_m}], \end{aligned} \tag{11}$$

with variational coefficients c_p and c_d, both initialised at zero. The three $2p_k$ terms, and the five $3d_m$ terms, enter with equal weights, so that the state is a ${}^1S$ singlet. Two electrons in the $2s$ and $2p$ orbitals can form ${}^1S^e$ only as $2s^2$ or $2p^2$, since $2s2p$ has odd parity. The first two terms of Eq. (11) are therefore the complete active space CAS(2,4) for this state. The third term adds the $3d^2$ configurations.

D. Jastrow factor

When two electrons coalesce, the cusp condition reads [9]

$$\left.\frac{\partial J}{\partial r_{ij}}\right|_{r_{ij}\to 0} = a_{ij}, \qquad a_{ij} = \begin{cases} 1/4 & \text{parallel spins,} \\ 1/2 & \text{antiparallel spins.} \end{cases} \tag{12}$$

The two values differ because for parallel spins D vanishes linearly at coalescence. A Padé term satisfies Eq. (12) for any range parameter,

$$J_{\mathrm{P}} = \sum_{i<j} \frac{a_{ij}\, r_{ij}}{1 + b_{ij}\, r_{ij}}, \tag{13}$$

with $b_{ij} = b_\parallel$ or $b_\perp$ according to the spins, bounded to $[0.05, 10]$.

The multi-configuration trial functions add a three-body term,

$$J = J_{\mathrm{P}} + \sum_{i<j} g_\theta(\mathbf{u}_{ij})\, e^{-0.3\,(t_i + t_j + t_{ij})}, \tag{14}$$

with $t_i = t(r_i)$, $t_{ij} = t(r_{ij})$, and the features

$$\mathbf{u}_{ij} = \big(t_i + t_j,\ t_i t_j,\ t_{ij},\ s_{ij},\ t_i t_j \cos\theta_{ij}\big), \tag{15}$$

TABLE I. Multi-configuration trial functions. "Modulation" gives the form of m_{nl}, Eq. (10); J_1 is the one-body term, Eq. (16). The last column is the seed of the optimisation.

Label	Configurations	Modulation	J_1	Param.	Seed
Q	$2s^2, 2p^2$	quadratic	no	396	42
Q′	$2s^2, 2p^2$	quadratic	no	396	7
L	$2s^2, 2p^2$	linear	no	393	42
U	$2s^2, 2p^2$	none	no	390	42
Q+J$_1$	$2s^2, 2p^2$	quadratic	yes	421	42
U+J$_1$	$2s^2, 2p^2$	none	yes	415	42
F+J$_1$	$2s^2, 2p^2$	$2s$, $2p$; $m_{1s} = 1$	yes	419	42
D	$2s^2, 2p^2, 3d^2$	quadratic	no	400	42
D$_U$	$2s^2, 2p^2, 3d^2$	none	no	392	42
D+J$_1$	$2s^2, 2p^2, 3d^2$	quadratic	yes	425	42

where $s_{ij} = \pm 1$ for parallel and antiparallel spins and θ_{ij} is the angle between $\mathbf{r}_i$ and $\mathbf{r}_j$. The features are symmetric under $i \leftrightarrow j$. Since they are built on t, the term does not change either cusp. g_θ is a multi-layer perceptron with two hidden layers of 16 units and the SiLU activation, $\sigma(x) = x/(1 + e^{-x})$, whose second derivative does not vanish. The same weights are used for all six pairs.

Some trial functions add a one-body term,

$$J_1 = \sum_i w_\theta\big(\ln[1 + t(r_i)]\big), \tag{16}$$

with w_θ a perceptron with one hidden layer of 8 SiLU units. J_1 has zero slope at the nucleus, so Eq. (4) still holds. At large r it grows at most as $\ln r$, so e^{J_1} is a power of r and Ψ stays normalisable.

E. The trial functions

a. Single-determinant series. The four trial functions share the determinant of Sec. II C and differ only in J. S0 has $J = 0$. S1 has the Padé term, Eq. (13). S2 replaces it by a network $f_\theta\big(r_{ij}/(1 + r_{ij}),\, s_{ij}\big)$, summed over pairs, which can represent a cusp of any slope but must find the values of Eq. (12) by optimisation. S3 adds to the Padé term a network acting on $t(r_{ij})/[1+t(r_{ij})]$; since $t'(0) = 0$, this network cannot change the slope at coalescence, and the cusps are those of the Padé term. The networks have two hidden layers of 16 units. The numbers of variational parameters are 3, 5, 340 and 342.

b. Multi-configuration trial functions. Table I lists the ten trial functions built on Eq. (11). Q is the reference: all modulations are quadratic. Q′ differs from Q only in the seed of the optimisation. L has linear modulations, U none. Q+J$_1$ and U+J$_1$ add the one-body term to Q and U. F+J$_1$ keeps $m_{1s} = 1$, modulates ϕ_{2s} and ϕ_{2p}, and has the one-body term. D, D$_U$ and D+J$_1$ add the $3d^2$ configurations to Q, U and Q+J$_1$. A parameter that is not used is kept at its initial value, $c^{(1)} = c^{(2)} = 0$.

III. STRUCTURE OF THE NODAL SURFACE

The Jastrow factor is positive, so the nodal surface of Ψ is the nodal surface of D_{CAS}. In this section we write D_{CAS} as the product of a positive function and a function that carries the node. We then list the transformations of the orbitals that leave the node unchanged, and the quantities that do not depend on them.

A. Factorisation

For an orbital χ, the spin-up determinant built from the pair (ϕ_{1s}, χ) can be written as

$$\phi_{1s}(r_0)\,\phi_{1s}(r_1)\,\big[\tilde\chi(\mathbf{r}_1) - \tilde\chi(\mathbf{r}_0)\big], \qquad \tilde\chi = \frac{\chi}{\phi_{1s}}, \tag{17}$$

and the spin-down determinant in the same way. We define two radial functions,

$$f(r) = \frac{\phi_{2s}(r)}{\phi_{1s}(r)} = (r - c)\, e^{(Z-\alpha_{2s})r}\, q_{2s}(r), \tag{18}$$

$$h(r) = \frac{\phi_{2p_k}(\mathbf{r})}{x_k\,\phi_{1s}(r)} = e^{(Z-\alpha_{2p})r}\, q_{2p}(r), \tag{19}$$

where $q_{nl} = m_{nl}/m_{1s}$. With $f_i = f(r_i)$ and $\mathbf{w}_i = h(r_i)\,\mathbf{r}_i$, Eq. (11) at $c_d = 0$ becomes

$$D_{\mathrm{CAS}} = \Phi\, N, \qquad \Phi = \prod_{i=0}^{3} \phi_{1s}(r_i), \tag{20}$$

$$\begin{aligned} N = {} & (f_1 - f_0)(f_3 - f_2) \\ & + c_p\,(\mathbf{w}_1 - \mathbf{w}_0)\cdot(\mathbf{w}_3 - \mathbf{w}_2). \end{aligned} \tag{21}$$

The sum over the three $2p_k^2$ configurations gives the scalar product. Φ is positive as long as m_{1s} has no zero; we checked this on a radial grid up to 15 bohr for every multi-configuration trial function. The nodal surface is the set $N = 0$. The function N depends on the electron positions only through f, h and c_p, id est through two radial functions and one number.

The $3d^2$ configurations add one term of the same form. Summing over the five configurations gives

$$\begin{aligned} N = {} & (f_1 - f_0)(f_3 - f_2) \\ & + c_p\,(\mathbf{w}_1 - \mathbf{w}_0)\cdot(\mathbf{w}_3 - \mathbf{w}_2) \\ & + c_d\,(W_1 - W_0) : (W_3 - W_2), \end{aligned} \tag{22}$$

with $W_i = h_d(r_i)\,T(\mathbf{r}_i)$, $h_d(r) = e^{(Z-\alpha_{3d})r}\, q_{3d}(r)$, and the colon denoting the full contraction of the two tensors. A constant factor from the normalisation of the $3d$ orbitals is absorbed in c_d.

B. Redundancies

Two sets of transformations leave the node unchanged. The first acts on f, h and c_p:

$$f \to af + b, \qquad h \to \mu h, \qquad c_p \to c_p \, \frac{a^2}{\mu^2}, \tag{23}$$

with constants $a \neq 0$, b and $\mu \neq 0$. Under Eq. (23), $N \to a^2 N$, so Ψ changes by a constant factor. The constant b corresponds to adding a multiple of ϕ_{1s} to ϕ_{2s}. A determinant does not change when a multiple of one column is added to another,

$$\det\big(\phi_{1s},\, \phi_{2s} + \lambda\phi_{1s}\big) = \det\big(\phi_{1s},\, \phi_{2s}\big), \tag{24}$$

so the orthogonalisation of ϕ_{2s} against ϕ_{1s} has no effect on Ψ, and we omit it. The constants a and μ are the scales of ϕ_{2s} and ϕ_{2p} relative to ϕ_{1s}. In the parametrisation of Sec. II these scales are not separate parameters, since $m_{nl}(0) = 1$; the radial parameters change them, over the region occupied by the electrons, together with the shape of the orbitals. The coefficient c_p therefore has no meaning on its own: two trial functions with nearly the same node can have different c_p. The same holds for c_d, with $W \to \nu W$ and $c_d \to c_d \, a^2/\nu^2$.

The second set acts on the modulations. If every m_{nl} is multiplied by the same positive function $g(r)$, D_{CAS} is multiplied by $\prod_i g(r_i)$. This is a one-body Jastrow factor. The functions f and h do not change, and neither does the node. The node depends on the modulations only through the ratios q_{nl}. Since α_{1s} is fixed at Z by the cusp condition, m_{1s} is the only radial freedom of the $1s$ orbital, and it enters the node only through these ratios.

C. Invariants

For a radius $r_\star$ we define

$$\hat{f}(r) = \frac{f(r) - f(0)}{f(r_\star) - f(0)}, \qquad \hat{h}(r) = \frac{h(r)}{h(r_\star)},$$
$$\tilde{c}_p = c_p \left[\frac{h(r_\star)}{f(r_\star) - f(0)}\right]^2. \tag{25}$$

These quantities do not change under Eq. (23), and

$$\frac{N}{\big[f(r_\star) - f(0)\big]^2} = (\hat{f}_1 - \hat{f}_0)(\hat{f}_3 - \hat{f}_2) + \tilde{c}_p \, (\hat{\mathbf{w}}_1 - \hat{\mathbf{w}}_0) \cdot (\hat{\mathbf{w}}_3 - \hat{\mathbf{w}}_2), \tag{26}$$

with $\hat{\mathbf{w}}_i = \hat{h}(r_i)\, \mathbf{r}_i$. The triple $(\hat{f}, \hat{h}, \tilde{c}_p)$ fixes the node. The number $\tilde{c}_p$ alone does not: trial functions with different $\hat{f}$ or $\hat{h}$ have different nodes at equal $\tilde{c}_p$, and the value of $\tilde{c}_p$ depends on $r_\star$. A comparison of $\tilde{c}_p$ between trial functions is meaningful only when their $\hat{f}$ and $\hat{h}$ are close over the region occupied by the electrons.

To measure how close they are, we match each trial function to a reference R. With $g = f - f(0)$ and a radial weight $\rho(r)$, the constants

$$a_\star = \arg\min_a \int dr\, \rho \, \frac{(ag - g_R)^2}{g_R^2},$$
$$\mu_\star = \arg\min_\mu \int dr\, \rho \, \frac{(\mu h - h_R)^2}{h_R^2} \tag{27}$$

bring f and h closest to those of R. The matched coefficient $\tilde{c}_p^R = c_p \, a_\star^2/\mu_\star^2$ is invariant under Eq. (23) and has no free radius. The relative residuals of the two fits, ϵ_f and ϵ_h, measure how far the radial functions of the trial function are from those of R.

We also use the weight of each configuration channel,

$$w_l = \frac{\int d\mathbf{R}\; e^{2J}\, \Phi^2\, (c_l\, T_l)^2}{\int d\mathbf{R}\; e^{2J}\, \Phi^2 \sum_{l'} (c_{l'}\, T_{l'})^2}, \qquad l \in \{s, p, d\}, \tag{28}$$

where T_s, T_p and T_d are the three products in Eq. (22) and $c_s = 1$. The integrand of the denominator has no node, so w_l is sampled without difficulty. The weights do not change under Eq. (23), since every channel scales by a^2. They depend on the positive factor $e^J\Phi$, which the node does not fix: two trial functions with the same node can have different w_l. The weights describe the trial function, not the node.

D. The limit $c_p = 0$

At $c_p = c_d = 0$, $N = (f_1 - f_0)(f_3 - f_2)$. If f is strictly monotonic in r, N vanishes exactly when $r_0 = r_1$ or $r_2 = r_3$. The node is then

$$\{r_0 = r_1\} \cup \{r_2 = r_3\}, \tag{29}$$

the node of the $1s^2 2s^2$ determinant [8], whatever the radial parameters. We checked that f is monotonic for every multi-configuration trial function. The four trial functions of the single-determinant series therefore share one node, and their energies differ only through the positive factor.

E. Sign disagreement as a probe of the node

A direct comparison of two nodal surfaces a and b is the probability that the two trial functions have opposite signs,

$$\varphi_{ab} = P_{|\Psi_a|^2}\big(\operatorname{sign}\Psi_a \neq \operatorname{sign}\Psi_b\big), \tag{30}$$

sampled from $|\Psi_a|^2$. This probe is weak. Near the node of Ψ_a, $|\Psi_a|^2 \approx |\nabla\Psi_a|^2 d^2$, with d the distance from the node. If the node of Ψ_b is displaced by $\delta(s)$ at the point s of the surface, the region between the two nodes carries the weight

$$\varphi_{ab} \approx \frac{1}{3} \, \frac{\oint dS\, |\nabla\Psi_a|^2\, \delta^3}{\int d\mathbf{R}\, |\Psi_a|^2}. \tag{31}$$

The probe is of third order in δ. The fixed-node energy changes to first order in δ for two generic nodes, and to second order about the exact node. A sample without sign disagreements bounds φ_{ab}, but it does not bound the difference of the fixed-node energies.

IV. METHODS

A. Variational Monte Carlo

We represent the trial function by $L = \ln|\Psi|$. The local energy is

$$E_L = -\frac{1}{2}\big(\nabla^2 L + |\nabla L|^2\big) - \sum_i \frac{Z}{r_i} + \sum_{i<j} \frac{1}{r_{ij}}, \qquad (32)$$

with the derivatives taken over the twelve coordinates by automatic differentiation. Configurations are sampled from $|\Psi|^2$ with 4000 Metropolis walkers. Each move displaces one electron by a Gaussian step; a sweep moves the four electrons in turn. Single-electron moves are used because the $1s$ and valence electrons have different length scales. The step is adapted to an acceptance of about 55%.

The parameters are optimised by stochastic reconfiguration (SR) [10]. With the scores $O_k = \partial_{\theta_k} \ln|\Psi|$, the gradient of the energy and the metric are

$$g_k = 2\,\big\langle (O_k - \langle O_k\rangle)(E_L - \langle E_L\rangle)\big\rangle, \qquad (33)$$

$$S_{kl} = \big\langle (O_k - \langle O_k\rangle)(O_l - \langle O_l\rangle)\big\rangle. \qquad (34)$$

The update is $\Delta\theta = -\eta\,(S + \gamma\,\mathrm{diag}\,S + 10^{-8})^{-1} g$, with $\gamma = 5\times10^{-3}$. The damping is relative to the diagonal of S because orbital parameters and network weights have different scales. The norm of $\Delta\theta$ is then limited to $\Delta_{\max}$. The learning rate and the trust radius start at $\eta_0 = 0.1$ and $\Delta_0 = 0.3$ and decrease together as $0.05 + 0.475\,[1 + \cos(\pi n / n_{\max})]$, with $n_{\max} = 3500$ epochs of ten sweeps each. In the gradient only, the local energies are clipped to five mean absolute deviations about their median. After the optimisation the energy is accumulated over 200 blocks separated by 150 sweeps, and its error is the largest standard error over the levels of a blocking analysis [11].

The single-determinant series was computed with an earlier version of the code. It used the same sampler and SR update, with γ between 10^{-3} and 5×10^{-3}, 300 (S0), 600 (S1) and 1500 (S2, S3) epochs, and 40 blocks separated by 150 sweeps.

B. Fixed-node diffusion Monte Carlo

FN-DMC samples the mixed distribution $\Psi\,\Phi_{\mathrm{FN}}$ with walkers that drift, diffuse and carry weights [4, 5]. A walker moves as $\mathbf{R}' = \mathbf{R} + \Delta\tau\,\bar{\mathbf{v}} + \boldsymbol{\eta}$, where $\boldsymbol{\eta}$ is Gaussian with variance $\Delta\tau$ per coordinate and $\bar{\mathbf{v}}$ is the drift $\nabla \ln|\Psi|$, reduced electron by electron by the factor $2/(1+\sqrt{1+2v^2\Delta\tau})$ of Ref. 12. The move is accepted with the Metropolis probability built on the drift-diffusion Green function. A move that changes the sign of Ψ is rejected; this imposes the node. The weight is multiplied by $\exp\{-\Delta\tau\,[(E_L(\mathbf{R}) + E_L(\mathbf{R}'))/2 - E_T]\}$, with E_L limited to $\pm 2/\sqrt{\Delta\tau}$ about a reference energy. Every five steps the weights are renormalised to a total of N_w, and the trial energy is set to $E_T = E_{\mathrm{run}} - \ln(W/N_w)/\tau_{\mathrm{fb}}$ with $\tau_{\mathrm{fb}} = 1$. When the effective sample size falls below $0.95\,N_w$ the population is resampled with a comb. The energy is the weighted average of E_L.

All lengths are given in imaginary time. The walkers start from the VMC walkers. After an equilibration of $\tau = 10$ the energy is accumulated in blocks of $\tau = 1$ over $\tau = 150$, or $\tau = 100$ for the trial functions with J_1. The time step is $\Delta\tau = 0.0025$ and $N_w = 4000$. Each production run has its own seed, except Q and U, which share one; the correlation of their blocks is $+0.04$, and we quote the error of their difference without correction. The error is obtained by blocking with a plateau test [11]. For every production run the plateau is reached at the first level: blocks of $\tau = 1$ are uncorrelated.

C. Nodal analysis

The functions f, h, $\hat{f}$, $\hat{h}$ and the coefficients $\tilde{c}_p$ and $\tilde{c}^R_p$ follow from the parameters without sampling. We take $r_\star = 2$ bohr unless stated otherwise. For Eq. (27) the reference is Q and ρ is the one-body radial density of Q, the same for every trial function. Its 5%, 50% and 95% quantiles are 0.14, 0.95 and 3.9 bohr. The weights w_l and the probabilities φ_{ab} are sampled with 4000 Metropolis chains over 40 blocks of 10 sweeps; their errors are the spread of the chain means divided by $\sqrt{4000}$.

D. Tests

We tested the FN-DMC code on the hydrogen atom with the nodeless trial function $e^{-1.3r}$, at four time steps from 0.02 to 0.0025. The linear extrapolation to $\Delta\tau = 0$ agrees with $-0.5\,E_h$ within 0.4 standard deviations. This test has no node, and does not test the fixed-node constraint. For Q we computed a time-step series, $\Delta\tau = 0.01$, 0.005 and 0.0025 with $\tau = 40$ each, and runs with 1000, 2000 and 4000 walkers with $\tau = 60$ each.

The code is written in JAX [13] with Flax, in double precision. A production run of $\tau = 150$ with 4000 walkers takes about 70 minutes on a laptop CPU.

V. RESULTS

A. Single-determinant series

Table II gives the energies of the series. The four trial functions have the same node (Sec. III D), so the differences come from the positive factor.

S0 lies 3.4 mHa above the Hartree-Fock limit. Its determinant has three radial parameters, and the Hartree-Fock limit is a lower bound for it. The Padé term, S1, lowers the energy by 53.6 mHa and the variance by a factor of 12.5. The network of S2 lowers the energy by a further 9.7 mHa and the variance by 4%. Adding the exact cusps to the network, S3, changes the energy by 0.2(1.6) mHa.

The network of S2 reproduces the antiparallel-spin cusp to 0.7% and gives the parallel-spin pairs a slope of 0.4529 instead of 0.25, an error of 81%. For parallel spins D vanishes at coalescence, so the region $r_{ij} \to 0$ carries little weight for these pairs, and only two of the six pairs have parallel spins. The gradient of the energy along the parallel-spin slope is therefore small. S3 returns the exact slopes to four decimals, as required by its construction.

Figure 1 shows the optimised Jastrow factors. In S1 the parallel-spin range parameter runs to the bound, $b_{\parallel} = 10$: the parallel-spin term becomes a constant, a/b, and is switched off. With the bound raised to 100, $b_{\parallel}$ runs again to the bound and the energy rises to $-14.5787\,E_h$. In S3, $b_{\parallel} = 0.283$, inside the interval. S1 and S3 have opposite parallel-spin correlation functions and the same energy within the error. In S2 the network grows linearly at large r_{ij}, with slopes 0.088 (antiparallel) and 0.07 (parallel). For an electron far from the others this gives a factor of about $e^{0.25r}$, which compensates part of the orbital decay; α_{2s} is 1.314 in S2 against 1.170 in S1. The network does one-body work, and the network and α_{2s} move along a direction of nearly constant energy.

B. Multi-configuration trial functions: VMC

Table III gives the VMC results. The energies lie between $-14.66620\,E_h$ (U) and $-14.66697\,E_h$ (D+J$_1$). The errors are 0.11–0.17 mHa, and about half of the differences between trial functions are below 1.5 standard deviations. The variance of the local energy separates them more: 0.0086 for Q+J$_1$ and D+J$_1$, 0.0095 for Q, 0.0131 for U.

Some coefficients end at a bound: $c^{(2)}_{1s} = 0.8$ in Q+J$_1$ and D+J$_1$, both coefficients of m_{3d} in D and D+J$_1$, and $c^{(1)}_{2s} = -0.15$ in F+J$_1$. For these trial functions the results depend on the bounds. In every trial function m_{1s} has no zero and f is monotonic. Equation (22) reproduces the directly computed determinant part to a relative accuracy of 10^{-14}.

In Q+J$_1$ the one-body term is negative: $w_\theta(r) - w_\theta(0) = -0.55$, -1.10 and -1.79 at $r = 1$, 2 and 4 bohr. In the same trial function m_{1s} grows more than in Q: $c^{(1)}_{1s} = 1.55$ against 1.30, and $c^{(2)}_{1s}$ reaches its bound. At $r = 4$ bohr, $\ln m_{1s} = 2.65$. The one-body term removes part of the one-body factor that m_{1s} carries. In U+J$_1$, where the orbitals are not modulated, the one-body term is positive (+0.51 at $r = 4$ bohr) and α_{2p} increases from 1.10 to 1.21.

C. Error budget

Table IV gives the FN-DMC energies. Figure 2 compares E_{VMC} and E_{FN} for Q, L and U. In every trial function $E_{\mathrm{VMC}} - E_{\mathrm{FN}}$ is larger than $E_{\mathrm{FN}} - E_{\mathrm{exact}}$. For Q, Q′, Q+J$_1$ and D+J$_1$ the positive factor accounts for 60–81% of the VMC error, for F+J$_1$ for 52% and for D for 98%. These fractions have errors of about 20 percentage points.

D. Radial modulation and the node

The fixed-node energy of Q is lower than that of U by 0.264(57) mHa, 4.7 standard deviations. The radial modulation changes the node. In VMC the difference is 0.577(179) mHa. Of this, 0.264 mHa comes from the node and 0.313(188) mHa from the positive factor. The VMC energy alone does not separate the two parts.

L lies between the two: $E_{\mathrm{FN}}(\mathsf{L}) - E_{\mathrm{FN}}(\mathsf{U}) = -0.196(55)$ mHa and $E_{\mathrm{FN}}(\mathsf{L}) - E_{\mathrm{FN}}(\mathsf{Q}) = +0.068(51)$ mHa. Q′, which differs from Q only in the seed of the optimisation, gives $E_{\mathrm{FN}}(\mathsf{Q}') - E_{\mathrm{FN}}(\mathsf{Q}) = +0.033(50)$ mHa. A second pair that differs in an element without effect on the node, U+J$_1$ and U (Sec. V G), gives $-0.072(72)$ mHa. In both cases the spread is consistent with the statistical error.

E. The node and the 1s orbital

Table V divides the nine trial functions with FN-DMC energies by construction: m_{1s} optimised or $m_{1s} = 1$. Within each group the fixed-node energies agree within their errors. The trial functions with $m_{1s} = 1$ lie 0.198(32) mHa higher. Figure 3 shows the nine energies.

The groups also differ in other elements: U and U+J$_1$ have no modulation of ϕ_{2s} and ϕ_{2p}. Three trial functions with the same Jastrow factor isolate the 1s orbital. From U+J$_1$ to F+J$_1$, ϕ_{2s} and ϕ_{2p} are modulated and ϕ_{1s} is not; the fixed-node energy changes by $-0.02(8)$ mHa. From F+J$_1$ to Q+J$_1$, m_{1s} is added; the fixed-node energy changes by $-0.20(6)$ mHa, 3.2 standard deviations. In VMC, Q+J$_1$ and F+J$_1$ differ by 0.14(17) mHa. Prima facie the two trial functions are equivalent.

The modulation of the 1s orbital acts on the node through the ratios q_{nl} of Sec. III B. With α_{1s} fixed at $Z = 4$ by the cusp condition, m_{1s} is the only way to change the shape of ϕ_{1s}. In Q and Q+J$_1$ it grows with r and makes ϕ_{1s} more diffuse than e^{-Zr}. Since $f = \phi_{2s}/\phi_{1s}$ and $h = \phi_{2p_k}/(x_k\phi_{1s})$, the shape of ϕ_{1s} enters the node. In F+J$_1$ the modulations of ϕ_{2s} and ϕ_{2p} cannot replace

TABLE II. Single-determinant series. Variational energy, variance of the local energy, and fraction of the correlation energy, computed with $E_{\mathrm{HF}} = -14.573023\,E_h$ [14] and $E_{\mathrm{exact}} = -14.667356\,E_h$ [7]. Bottom: slopes of J at coalescence compared with Eq. (12).

	S0	S1	S2	S3
Parameters	3	5	340	342
E (E_h)	−14.5696(45)	−14.6232(11)	−14.6329(11)	−14.6331(12)
Var(E_L)	3.208	0.256	0.246	0.250
Correlation	−3.6%	53.2%	63.5%	63.7%
$\partial J/\partial r_{ij}\|_0$, parallel (1/4)	–	0.2500	0.4529	0.2500
$\partial J/\partial r_{ij}\|_0$, antiparallel (1/2)	–	0.5000	0.4964	0.5000

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FIG. 1. Optimised Jastrow exponents of S1, S2 and S3, shifted to $J(0) = 0$. Solid lines: antiparallel pairs; dashed lines: parallel pairs. Shaded histogram: distribution of the sampled r_{ij} (right axis). Right panel: the coalescence region, with the exact cusp slopes 1/4 and 1/2 as dotted lines.

TABLE III. VMC energies of the multi-configuration trial functions, variance of the local energy, fraction of the correlation energy, and mixing coefficient c_p.

	E_{VMC} (E_h)	Var(E_L)	Corr. (%)	c_p
Q	−14.66677(12)	0.0095	99.38(13)	−0.2200
Q′	−14.66669(17)	0.0109	99.29(18)	−0.2226
L	−14.66660(12)	0.0113	99.19(13)	−0.1984
U	−14.66620(13)	0.0131	98.77(14)	−0.2437
Q+J$_1$	−14.66689(11)	0.0086	99.51(11)	−0.2184
U+J$_1$	−14.66633(14)	0.0122	98.91(15)	−0.2372
F+J$_1$	−14.66676(14)	0.0115	99.36(14)	−0.1904
D	−14.66663(12)	0.0107	99.23(13)	−0.2387
D$_{\mathrm{U}}$	−14.66647(13)	0.0134	99.06(14)	−0.2569
D+J$_1$	−14.66697(12)	0.0086	99.59(13)	−0.2376

TABLE IV. FN-DMC energies at $\Delta\tau = 0.0025$, distance from the exact energy, difference from the VMC energy, and fraction of the VMC error due to the positive factor, $(E_{\mathrm{VMC}} - E_{\mathrm{FN}})/(E_{\mathrm{VMC}} - E_{\mathrm{exact}})$. Energy differences in mHa.

	E_{FN} (E_h)	$E_{\mathrm{FN}} - E_{\mathrm{ex}}$	$E_{\mathrm{VMC}} - E_{\mathrm{FN}}$	Fraction
Q	−14.667237(37)	+0.119(37)	0.46(13)	80%
Q′	−14.667204(33)	+0.152(33)	0.52(17)	77%
L	−14.667169(35)	+0.187(35)	0.57(13)	75%
U	−14.666973(43)	+0.383(43)	0.78(14)	67%
Q+J$_1$	−14.667269(41)	+0.087(41)	0.38(11)	81%
U+J$_1$	−14.667045(58)	+0.311(58)	0.71(15)	70%
F+J$_1$	−14.667066(49)	+0.290(49)	0.31(15)	52%
D	−14.667340(69)	+0.016(69)	0.71(14)	98%
D+J$_1$	−14.667201(35)	+0.155(35)	0.23(13)	60%

it: $c_{2s}^{(1)}$ ends at its lower bound. In Q+J$_1$ the coefficient $c_{1s}^{(2)}$ is at its upper bound, so the size of the effect may depend on the bound.

TABLE V. Fixed-node energies grouped by the treatment of the 1s orbital. Weighted means, in mHa above the exact energy, and χ^2 about the mean.

Group	Trial functions	$E_{\mathrm{FN}} - E_{\mathrm{ex}}$	χ^2/dof
m_{1s} optimised	Q, Q′, L, Q+J$_1$, D, D+J$_1$	+0.137(16)	7.3/5
$m_{1s} = 1$	U, U+J$_1$, F+J$_1$	+0.335(28)	2.3/2
Difference		0.198(32)	

TABLE VI. Mixing coefficient c_p, invariant $\tilde{c}_p$ at $r_\star = 2$ bohr, matched coefficient $\tilde{c}_p^R$ (Eq. (27), reference Q), and relative residuals of the match.

	c_p	$\tilde{c}_p$	$\tilde{c}_p^R$	ϵ_f	ϵ_h
Q	−0.2200	−0.1078	−0.2200	0	0
Q′	−0.2226	−0.1084	−0.2211	0.024	0.024
L	−0.1984	−0.1060	−0.2115	0.156	0.168
U	−0.2437	−0.0998	−0.1936	0.639	0.607
Q+J$_1$	−0.2184	−0.1081	−0.2121	0.043	0.030
U+J$_1$	−0.2372	−0.1001	−0.1947	0.547	0.491
F+J$_1$	−0.1904	−0.1058	−0.2174	0.483	0.460
D	−0.2387	−0.1091	−0.2301	0.062	0.071
D$_{\mathrm{U}}$	−0.2569	−0.1015	−0.1888	0.629	0.586
D+J$_1$	−0.2376	−0.1080	−0.2212	0.063	0.071

F. Redundancies in the optimised parameters

Table VI gives, for every trial function, c_p, $\tilde{c}_p$ at $r_\star = 2$ bohr, the matched coefficient $\tilde{c}_p^R$ with Q as reference, and the residuals of the match. Figure 4 shows the modulations, the ratios q_{nl} and the functions $\hat{f}$, $\hat{h}$ for Q, Q′, L and U.

Among the nine trial functions with FN-DMC energies the raw coefficient c_p ranges from −0.190 to −0.244. It does not order the fixed-node energies: a linear fit of E_{FN} in $|c_p|$ gives $\chi^2 = 46.8$ for 7 degrees of freedom. The

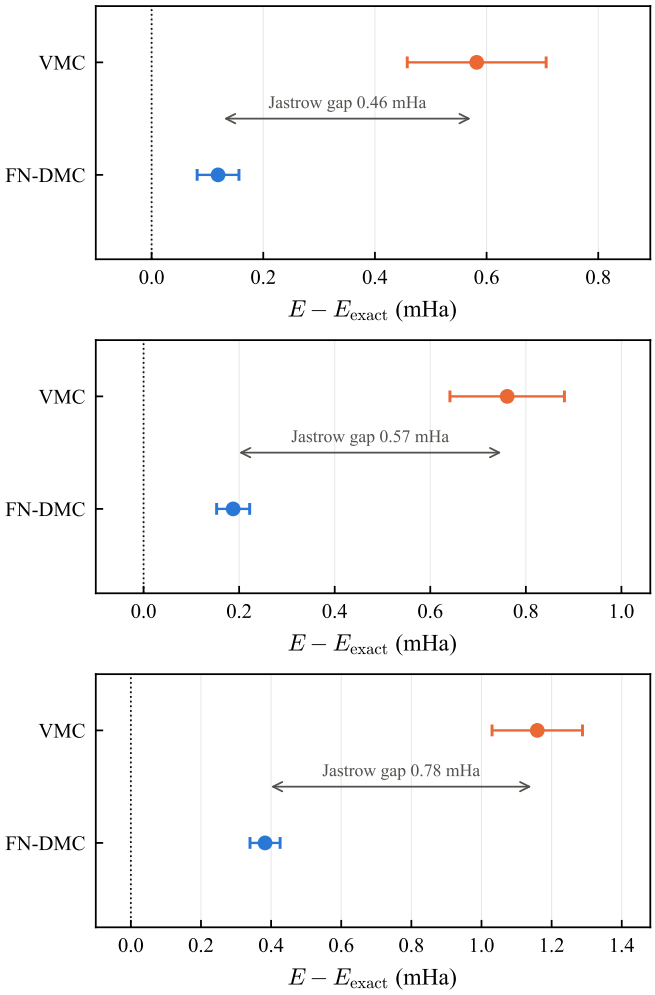


FIG. 2. VMC and FN-DMC energies relative to the exact energy for Q (top), L (middle) and U (bottom). The arrow is $E_{\text{VMC}} - E_{\text{FN}}$, the error of the positive factor.

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FIG. 3. Fixed-node energies of the nine trial functions relative to the exact energy, grouped by the treatment of the $1s$ orbital. Bands: weighted means of Table V.

five trial functions whose radial functions lie within 7.1% of those of Q (Q, Q′, Q+J_1, D, D+J_1) have c_p between -0.218 and -0.239 and fixed-node energies that agree within their errors ($\chi^2 = 4.8/4$). For these trial functions the spread of $\tilde{c}_p^R$ is 8%.

The trial functions with $m_{1s} = 1$ have radial functions 46–64% away from those of Q. For them a single number cannot describe the node. Table VII shows the consequence: a linear fit of E_{FN} in $|\tilde{c}_p|$ over the nine trial functions has a slope that changes sign with $r_\star$. At small $r_\star$ the trial functions with $m_{1s} = 1$ have the largest $|\tilde{c}_p|$,

TABLE VII. Linear fit of $E_{\text{FN}} - E_{\text{ex}}$ in $|\tilde{c}_p|$ for the nine trial functions, at several gauge-fixing radii and with the matched coefficient. Slope in units of its standard error.

Gauge	Slope (σ)	χ^2/dof
$r_\star = 0.5$ bohr	+5.7	14.1/7
$r_\star = 1.0$ bohr	+5.3	18.9/7
$r_\star = 1.5$ bohr	−3.0	37.9/7
$r_\star = 2.0$ bohr	−6.2	8.7/7
$r_\star = 3.0$ bohr	−5.6	16.4/7
Matched to Q	−5.4	17.8/7

at large $r_\star$ the smallest. The fit separates the two groups of Table V in either direction. The matched coefficient gives $\chi^2 = 17.8/7$, and no slope within the five modulated trial functions without $3d^2$ configurations.

The coefficient of the $3d^2$ configurations behaves in the same way. From D_U to D, $|c_d|$ decreases from 0.078 to 0.016, while the weight of the channel, Eq. (28), increases from 6.8×10^{-5} to 1.05×10^{-4}. The change of c_d comes from the scale of the $3d$ orbital: in D, $\alpha_{3d} = 2.02$ and both coefficients of m_{3d} are at their bounds.

The VMC energy does not order the fixed-node energies either. The rank correlation of E_{VMC} and E_{FN} over the nine trial functions is 0.57 ($p = 0.11$). For the variance of the local energy it is 0.81 ($p = 0.008$).

G. The $3d^2$ configurations and the one-body term

Two pairs of trial functions differ only in the $3d^2$ configurations: $E_{\text{FN}}(\text{D}) - E_{\text{FN}}(\text{Q}) = -0.103(78)$ mHa and $E_{\text{FN}}(\text{D+J}_1) - E_{\text{FN}}(\text{Q+J}_1) = +0.068(54)$ mHa. The weighted mean is $+0.01(4)$ mHa. The $3d^2$ configurations do not change the fixed-node energy within 0.04 mHa, in agreement with a channel weight of 10^{-4}. D lies 1.8 standard deviations below the mean of the other five trial functions with optimised m_{1s}. Its error is twice that of the other runs of the same length. The standard deviation of its block energies is 0.84 mHa, against 0.35–0.46 mHa for Q, Q′ and D+J_1. Its excess kurtosis, 1.5, lies within the range of the other runs (-0.1 to 2.8), so the width does not come from a few outlying blocks. The blocks are uncorrelated, and the error is the standard error of the block mean.

Three pairs differ only in the one-body term: $E_{\text{FN}}(\text{Q+J}_1) - E_{\text{FN}}(\text{Q}) = -0.032(56)$ mHa, $E_{\text{FN}}(\text{U+J}_1) - E_{\text{FN}}(\text{U}) = -0.072(72)$ mHa and $E_{\text{FN}}(\text{D+J}_1) - E_{\text{FN}}(\text{D}) = +0.139(77)$ mHa. The weighted mean is 0.00(4) mHa. In VMC the one-body term lowers the energy of U by 0.14(19) mHa and that of Q by 0.12(16) mHa. It lowers the variance of Q and U by 7–10% and that of D by 20%.

H. Sign disagreement

Among Q, Q′, L and U, no pair has a sign disagreement in the sampled configurations. With 4000 independent

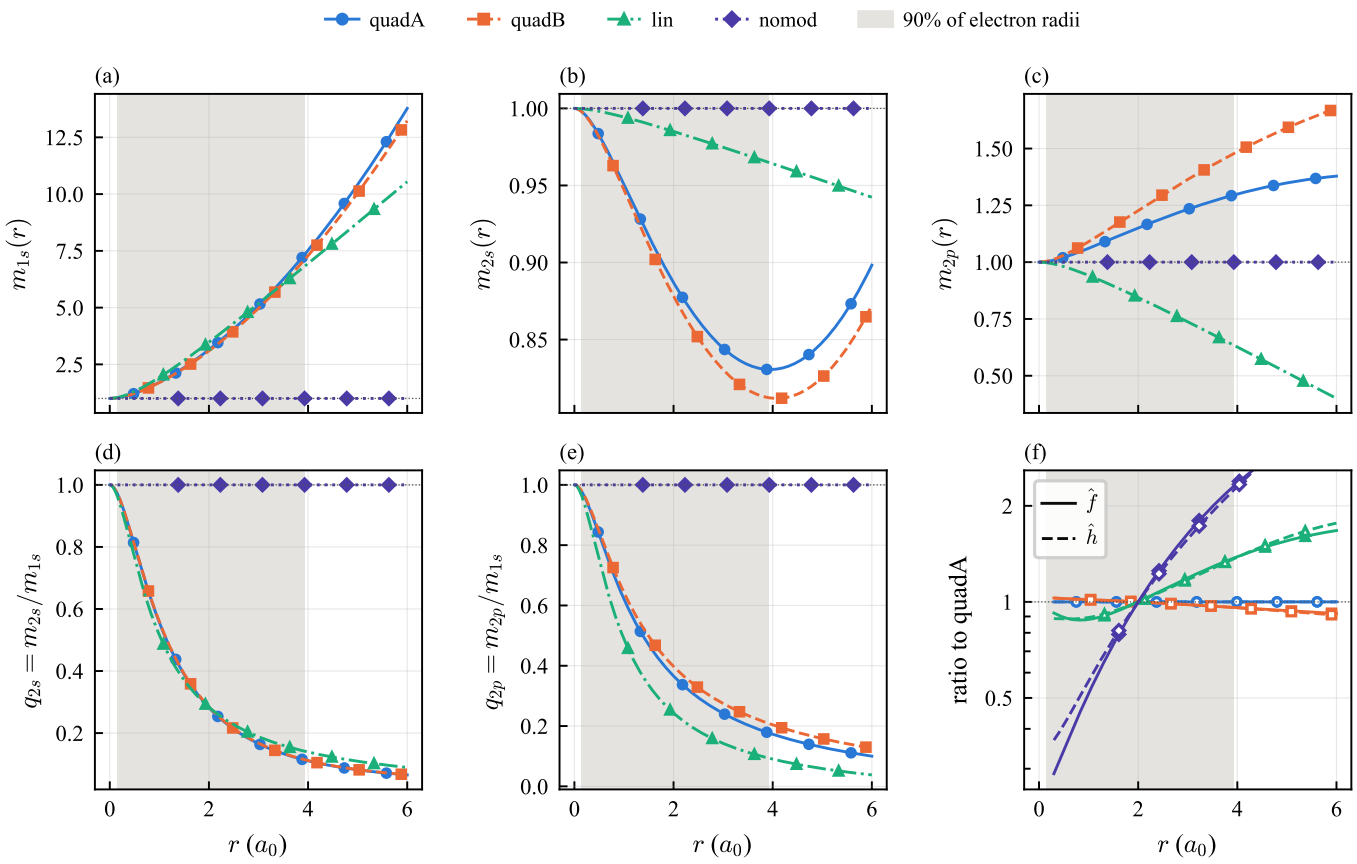


FIG. 4. Radial functions of Q, Q′, L and U. (a)–(c) Modulations m_{1s}, m_{2s}, m_{2p}. (d), (e) Ratios q_{2s} and q_{2p}, which enter the node. (f) $\hat{f}$ and $\hat{h}$ at $r_\star = 2$ bohr, as ratios to those of Q; by construction the ratios equal one at $r_\star$. Shaded band: central 90% of the sampled electron radii.

chains this gives $\varphi_{ab} < 7.5 \times 10^{-4}$ at 95% confidence. The fixed-node energies of Q and U differ by 0.264(57) mHa. Both facts follow from Eq. (31): the probe is of third order in the displacement of the node, the energy of first order.

I. Systematic errors

Table VIII and Fig. 5 give the time-step series of Q. The data do not fix the form of the time-step error. A linear and a quadratic fit extrapolate to +0.19 and +0.15 mHa above the exact energy; a constant fits the data with $\chi^2 = 6.8/3$. The production value at $\Delta\tau = 0.0025$ is +0.119(37) mHa. A linear or quadratic form places the error at $\Delta\tau = 0.0025$ between −0.05 and −0.01 mHa. We quote all fixed-node energies at $\Delta\tau = 0.0025$, with this range as a systematic uncertainty. The series exists only for Q; for the differences between trial functions we assume that the time-step error at $\Delta\tau = 0.0025$ is the same within this range. The fraction of rejected node crossings is 0.0169%, 0.0059% and 0.0020% at the three time steps, and 0.0020–0.0022% in every production run.

With 1000, 2000 and 4000 walkers the energies of Q are −14.667378(107), −14.667182(81) and −14.667201(53) E_h. A constant fits them with $\chi^2 = 2.5/2$; a term in $1/N_w$ has a slope of −0.21(16) mHa at $N_w = 1000$. The population error at $N_w = 4000$ is below the statistical error.

TABLE VIII. Time-step series of Q (top) and fits to it (bottom). $E_{\mathrm{FN}} - E_{\mathrm{ex}}$ in mHa.

$\Delta\tau$	$E_{\mathrm{FN}} - E_{\mathrm{ex}}$	Crossings
0.01	−0.046(64)	0.0169%
0.005	+0.154(63)	0.0059%
0.0025	+0.152(65)	0.0020%
0.0025, production	+0.119(37)	0.0020%
Fit	Intercept	χ^2/dof
Constant	+0.103(26)	6.8/3
Linear	+0.193(48)	1.7/2
Quadratic	+0.150(33)	1.0/2

J. Comparison with previous work

Table IX compares our results with those of Seth *et al.* [15] and Morales *et al.* [16]. The VMC energy of D+J$_1$

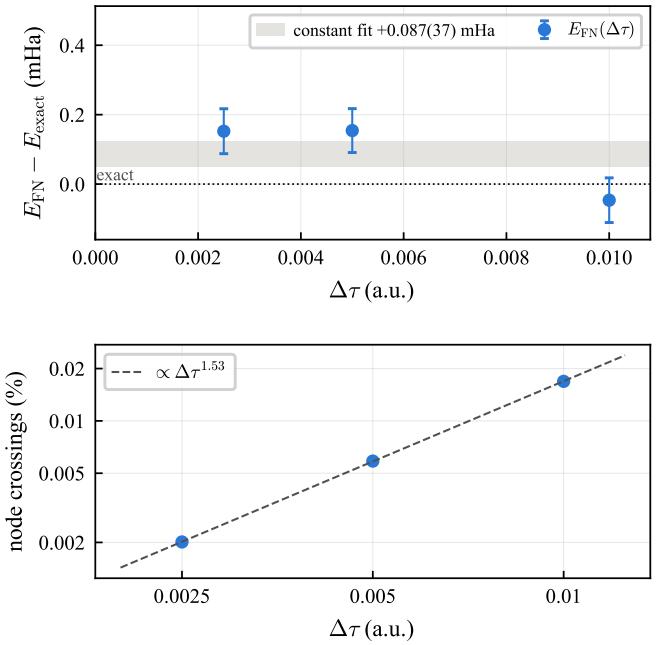


FIG. 5. Time-step series of Q. (a) $E_{\mathrm{FN}} - E_{\mathrm{exact}}$ and fits. (b) Fraction of rejected node crossings, logarithmic scales.

TABLE IX. Comparison with previous work. Energies in E_h.

	E_{VMC}	E_{FN}
This work, D+J_1	−14.66697(12)	−14.667201(35)
This work, mean, m_{1s} opt.	–	−14.667219(16)
Seth *et al.* [15]	−14.66719(1)	−14.667306(7)
Morales *et al.* [16]	−14.66688(4)	–
Exact [7]	−14.667356	

lies 0.22(12) mHa above that of Seth *et al.*, who use a larger expansion and backflow, and agrees with that of Morales *et al.*. The variance of D+J_1, 0.0086, lies between the values of Seth *et al.* without backflow, 0.0107, and with backflow, 0.0053. The fixed-node energy of the trial functions with optimised m_{1s} lies 0.087(17) mHa above that of Seth *et al.* Our value is not extrapolated in $\Delta\tau$. The extrapolation would raise it by up to 0.05 mHa.

VI. DISCUSSION

The network learns the parts of the Jastrow factor to which the energy is sensitive. In the single-determinant series the network reproduces the antiparallel-spin cusp and not the parallel-spin cusp, because the energy depends on the first and hardly on the second. Imposing both cusps costs one change of variable and gives the same energy. In the multi-configuration trial functions the positive factor still carries most of the VMC error: $E_{\mathrm{VMC}} - E_{\mathrm{FN}}$ is 0.2–0.8 mHa against a fixed-node error of 0.02–0.38 mHa.

The Jastrow factor cannot learn the node, since $e^J > 0$. The node is fixed by the determinant part, and in the trial functions studied here by two radial functions and one number. The parameters of the determinant part, optimised together with the Jastrow factor, are not identifiable: c_p and c_d change with the scale of the orbitals, and a common radial factor of the orbitals is a one-body Jastrow factor. What changes the node is the shape of each orbital relative to the $1s$ orbital. In our parametrisation the cusp condition fixes the $1s$ exponent, and the modulation m_{1s} is the radial freedom that the node needs.

Neither the VMC energy nor the raw coefficients detect a change of the node. F+J_1 and Q+J_1 have the same VMC energy within 0.14(17) mHa and fixed-node energies 0.20(6) mHa apart. A gauge-invariant coefficient detects the change only between trial functions with the same radial functions; across different radial functions its ordering depends on the choice of gauge. To compare nodes of different radial form we used FN-DMC. The variance of the local energy correlates with the fixed-node energy better than the VMC energy, but it also depends on the positive factor.

VII. LIMITATIONS AND OUTLOOK

The study treats one atom with a quasi-degenerate ground state. The effects on the node are 0.1–0.3 mHa, a few standard deviations of our fixed-node energies. Only one pair of trial functions, Q and Q′, differs only in the seed of the optimisation. The time-step series and the population test exist only for Q. The FN-DMC code was tested only on a nodeless system. Some coefficients of the best trial functions end at their bounds. The single-determinant series was computed with an earlier version of the code and a shorter protocol.

The invariants of Sec. III hold for the CAS(2,4) expansion with the orbitals of Sec. II. A measure of the distance between two nodes that does not depend on a choice of gauge would allow nodes of different radial form to be compared without FN-DMC. A parametrisation of the ratios q_{nl} instead of the modulations m_{nl} would remove the redundancy of Sec. III B and the bounds that now limit the $1s$ orbital. The same analysis applies to other atoms with a $2s^2$–$2p^2$ near-degeneracy.

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